

BoxFit

BoxFit is an enumeration that describes how a box should be inscribed into another box. It is commonly used in Flutter to control how images are scaled and positioned within their containers.



BoxFit Options

1- BoxFit.fill

Stretches the image to fill the container (aspect ratio is ignored).

```
Image.asset( 'assets/image.png', fit: BoxFit.fill, )
```

2- BoxFit.contain

Fits the whole image inside the container while keeping the aspect ratio.

```
Image.asset( 'assets/image.png', fit: BoxFit.contain, )
```

3- BoxFit.cover

Covers the entire container, keeps aspect ratio, but may crop parts.

```
Image.asset( 'assets/image.png', fit: BoxFit.cover, )
```

4- BoxFit.fitWidth

Scales image to match container's width (may overflow in height).

```
Image.asset( 'assets/image.png', fit: BoxFit.fitWidth, )
```

More BoxFit Options

5- BoxFit.fitHeight

Scales image to match container's height (may overflow in width).

```
Image.asset( 'assets/image.png', fit: BoxFit.fitHeight, )
```

6- BoxFit.none

No scaling, image keeps its original size.

```
Image.asset( 'assets/image.png', fit: BoxFit.none, )
```

7- BoxFit.scaleDown

Same as contain but only scales down if image is larger than container.

```
Image.asset( 'assets/image.png', fit: BoxFit.scaleDown, )
```

TextFormField Borders

1- UnderlineInputBorder

- This border draws a **single line** under the TextFormField.
- It is the **default border style** when you create a TextFormField.
- Commonly used in **material design forms** for a simple and minimal look.

```
TextFormField( decoration:  
  InputDecoration( border:  
    UnderlineInputBorder(),  
    labelText: 'Underline Border', ),  
  )
```

2- OutlineInputBorder

- This border draws a **full rectangular outline** around the TextFormField.
- It gives a more **defined and boxed look** compared to the underline border.
- Often used when you want the input fields to be **visually separated** and more noticeable.

```
TextFormField( decoration:  
  InputDecoration( border:  
    OutlineInputBorder(), labelText:  
    'Outline Border', ), )
```

3- InputBorder.none

- This completely **removes the border** from the TextFormField.
- Useful when you want a **clean, borderless design**, for example inside custom containers or cards.

```
TextFormField( decoration:  
  InputDecoration( border:  
    InputBorder.none, labelText:  
    'No Border', ), )
```

Border States

4- enabledBorder

- The border style that appears when the TextFormField is **enabled but not focused**.
- Usually displayed with a **neutral color** (grey).
- This helps indicate that the field is available for input, but the user is not currently typing in it.

```
TextFormField( decoration:
InputDecoration(
enabledBorder:
OutlineInputBorder( borderSide:
BorderSide(color: Colors.grey,
width: 1), ), labelText: 'Enabled
Border', ), )
```

5- focusedBorder

- The border style that appears when the TextFormField is **active and being typed into**.
- Typically shown with a **highlighted color** (like blue) to grab the user’s attention.
- This state visually tells the user which field they are interacting with.

```
TextFormField( decoration:
InputDecoration(
focusedBorder:
OutlineInputBorder( borderSide:
BorderSide(color: Colors.blue,
width: 2), ), labelText: 'Focused
Border', ), )
```

6- errorBorder

- The border style shown when the TextFormField has **invalid input** (failing validation).
- Usually displayed in **red** to clearly indicate an error.
- Appears only when an error message is triggered.

```
TextFormField( decoration:
InputDecoration( errorBorder:
OutlineInputBorder( borderSide:
BorderSide(color: Colors.red,
width: 2), ), labelText: 'Error
Border', ), )
```

7- focusedErrorBorder

- A combination of **focused** and **error** states.
- It appears when the field is in **error state** but also **currently focused**.
- Typically a brighter red (e.g., redAccent) to highlight that the user is typing into a field with invalid input.

```
TextFormField( decoration: InputDecoration(
focusedErrorBorder: OutlineInputBorder( borderSide:
BorderSide(color: Colors.redAccent, width: 2), ),
labelText: 'Focused Error Border', ), )
```

8- disabledBorder

- The border style shown when the TextFormField is **disabled** (enabled: false).
- Usually displayed in **grey or lighter colors** to indicate that the field cannot be interacted with.
- This provides a clear visual distinction between active and inactive fields.

```
TextFormField( decoration: InputDecoration(
disabledBorder: OutlineInputBorder( borderSide:
BorderSide(color: Colors.grey, width: 1), ), labelText:
'Disabled Border', ), enabled: false, )
```